Kamil Tollinger



Senior Backend Software Engineer · Tech Lead · Technical Analyst

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Profile

Senior software engineer with over 20 years of experience building critical business systems, primarily in the banking and financial sector. Focused on server-side development in Java and microservices architecture, with a strong overlap into analysis, software architecture and quality assurance. Actively using modern AI tools to increase development efficiency and code quality.

- Strong analytical and problem solving skills
- Quick learner who enjoys keeping current with new projects
- Ability to work independently and as a team member
- Handle crisis situations in calm and capable manner
- Ability to use multiple programming languages and operating environments
- Experience with financial sector (e-banking)

Selected projects

MultiSigning, Komerční banka, a.s. (2021–2026)

MultiSigning is a large microservices-based application responsible for orchestrating the signing of documents. A client system creates a signing case, registers documents and the application then guides all participants through the necessary steps up to final signature.

- Developing microservices (Java, Spring Boot) using an architecture based on Domain-Driven Design (DDD)
- Implementing backend layers: Business Components (BC) connected to PostgreSQL, Redis, and Kafka, alongside Backend for Frontend (BFF) integrating with an Angular SPA
- Collaboration on internal shared/core libraries in Kotlin (code reviews, RFCs, and API design requirements) and integrating them into Java 25 / Spring Boot microservices
- Securing API endpoints using JWT tokens (technical and user) and OAuth Proxy integration
- Managing infrastructure and containers in an OpenShift / Kubernetes (PaaS) environment using Helm/YAML manifests

Knowledge summary

Languages

- Czech - native
- English - fluent

Backend and integration technologies

- Java (JDK 8–25)
- Kotlin (Quarkus backend, library integration)
- Spring / Spring Boot (including Spring Batch)
- Quarkus (REST backends and microservices)
- REST (Jersey), SOAP, WS-Security
- Security: OAuth 2, JWT tokens
- Kafka, Redis
- Microservices architecture, Kubernetes
- Architecture: Domain-Driven Design (DDD), BFF (Backend for Frontend) pattern

Frontend and web technologies

- HTML, CSS, JavaScript, TypeScript
- Vue 3, Pinia
- VitePress (documentation)

Databases and persistence

- PostgreSQL, Oracle, MySQL, SQL

- Advanced CI/CD pipelines (Jenkins, Bitbucket), executing deployments via GitOps (ArgoCD), and troubleshooting via the K8s dashboard
- Applying agile practices and Trunk Based Development, including the automatic provisioning of dynamic Review environments in K8s for every Pull Request
- Designing and implementing automated post-deployment regression tests (TypeScript / Mocha)

Personal finance web application (side project, 2023–2026)

In parallel to my full-time job I am developing a small web application for personal accounting and budget planning. The application has several screens and works with about ten tables in the database. Currently undergoing progressive backend refactoring and migration from Java to Kotlin.

- Role: main author and solution architect
- Backend: PostgreSQL, Java 25 → progressive migration to Kotlin, Quarkus 3, REST API, JWT authentication
- Frontend: TypeScript, Vue 3, Pinia
- Documentation: VitePress
- Goal: learn modern technologies (transitioning to Kotlin, Quarkus) and build a practical tool for managing personal finances

Financial planner – Android application (2024)

A simple Android application with several screens for monthly financial planning.

- Role: main author and solution architect
- Technologies: Android SDK 14, Kotlin 2.0, Room, Jetpack Compose
- Goal: learn the modern Kotlin/Android stack and create a practical tool for personal financial planning

CESS/CPT, Česká spořitelna, a.s. (2016-2021)

Customer Pricing Tool is configuration database, set of tools for managing configuration data and many (~70) Spring Batch jobs/applications for evaluating client fees and benefits.

- Member of Agile (Scrum) team - Role: developer, technical support, production support
- Technologies: Java, Spring, Spring Batch, REST, SOAP, Oracle SQL
- JDBC, JPA

Programming languages and platforms

- Java (primary language, JDK 8–25)
- Kotlin (code reviews and integration of shared libraries in production, Android apps, backend migration)

XML and related technologies

- XML, XML Schema, XPath, XSLT (including generating PDF from XML)

DevOps, tools and version control

- Git, Trunk Based Development
- Jenkins, Bitbucket, Nexus (Maven Artifact Repository)
- GitOps (ArgoCD), Helm
- OpenShift (PaaS)
- Atlassian Jira (user and administration experience)

Testing and quality

- JUnit, Mockito, EasyMock
- Test Driven Development (TDD)

AI-assisted development

- Active use of tools like GitHub Copilot / Windsurf for development and refactoring
- Ability to craft precise prompts and guide larger refactorings with AI while keeping the code readable and consistent

Cryptography

- XML Signature, PKCS#7, PKCS#12, PKI

Operating systems

- Unix / Linux - installation / administration

Cloud Data Backup Tool, Česká spořitelna, a.s. (2015)

Spring Boot CLI tool for incremental backup of user data from a Google Cloud domain. Responsible for architecture, data model and implementation (Java, Spring Boot, Google Cloud Services).

SyDesk, StringData, s.r.o. (2007-2016)

SyDesk is application and infrastructure monitoring system. Low-level agents collects data across infrastructure, data are collected in database and processed. There are two major outputs/goals: notifications about actual or oncoming problem and generating (SLA) reports.

- Responsibilities: Agile Team Leader (Kanban), Architect, Developer
- Technologies: Java, PHP, Oracle SQL

Contract Archive, Česká Spořitelna (2014-2015)

Electronic Contract Archive – workflow web application for internal documents in the Google Cloud environment. Team lead, analysis and development (Java, REST, Google AppEngine, Google Datastore, Cloud SQL, AngularJS).

Brasil/CSKN, Česká Spořitelna (2012)

Part of project Brasil – unified front-end for the branch network and introduction of SOA architecture within the bank. Developed a module for parsing cadastral XML data and generating printable PDF documents (Java, XML, XSLT).

ELBST, ČSOB, a.s. (2010)

Workflow-driven web application supporting UAT tests and communication between internal and external teams. Analysis, development and support (Java, JSP, Oracle SQL).

Earlier projects (1994–2007)

Electronic banking system OfficeLine used by about fifteen banks – entering payment orders and secure transfer to the bank with strong encryption and digital signing (team lead, development, implementation, integration of banking systems).

Development of encryption and signing libraries for banking systems (DES, RSA), used as part of secure transfer workflows between backend systems.

Methodology

- Agile Development (Extreme Programming, Scrum)

Educations

- 1984-1988 Střední průmyslová škola strojní a elektrotechnická obor Elektrotechnika.

Trainings

- Developing Solutions for Microsoft Azure (AZ-204) – in progress (Udemy)
- Microsoft Certified: Azure Fundamentals (AZ-900) – in progress (Udemy)
- Kotlin for Java Developers (Coursera / JetBrains 2026)
- Security Maturity, level I (Komerční banka 2023)
- Kotlin for Java Developers (Udemy 2023)
- VueJS - The Complete Guide (Udemy 2023)
- Android Development Masterclass (Udemy 2021)
- Oracle Java Certification (Udemy 2017)
- AngularJS (Angular.cz 2015)
- JavaEE 6 Application Development (GOPAS 2015)